Data Cleaning & Preprocessing Report

ApexLark AI & Machine Learning Internship — Day 2

Project objective

Prepare a real-world dataset for machine-learning use by inspecting its structure, checking data quality, handling duplicates and missing values, transforming variables, detecting outliers, validating the result, and communicating business observations.

1. Dataset Collection

Dataset: UCI Machine Learning Repository — Wine Quality. The red wine file contains 1,599 records and the white wine file contains 4,898 records. The two files are combined, and a wine_type field is added to distinguish red and white samples. The combined source therefore contains 6,497 records before duplicate removal. The dataset includes physicochemical measurements and a quality score.

Dataset source: https://archive.ics.uci.edu/dataset/186/wine+quality

2. Dataset Inspection

The notebook displays the first and last ten records, shape, column names, data types through info(), and descriptive statistics through describe(). These checks help understand the dataset before modifying it.

3. Data Cleaning

- Missing values are identified with isnull().sum(). Numeric missing values are filled with the column median; categorical missing values are filled with the mode.
- Duplicate rows are counted and removed with duplicated() and drop_duplicates().
- Column names are standardized to lowercase with underscores. Text values in wine_type are stripped of extra spaces and converted to lowercase.
- Numeric columns are converted with pandas.to_numeric(errors='coerce').
- Constant columns are checked and removed if present.

4. Data Preprocessing

- Label Encoding converts the wine_type category into a numeric code.
- The Interquartile Range (IQR) method identifies possible outliers using Q1 – 1.5 × IQR and Q3 + 1.5 × IQR.
- Numeric predictor values are capped at their IQR bounds. The quality score and encoded category are not capped.
- Two derived features are created: total_acidity (fixed acidity + volatile acidity + citric acid) and alcohol_sugar_ratio (alcohol divided by residual sugar, with a small epsilon to avoid division by zero).
- The notebook checks for date columns and converts any available date field to DateTime. This dataset does not normally include a date column.

5. Data Validation

The notebook reports total missing values, duplicate rows, final dataset shape, updated data types, and counts of categorical and numerical features. The cleaned dataset is saved as outputs/cleaned_wine_quality.csv. Exact final counts are generated when the notebook is run, because they depend on the dataset version and cleaning results.

6. Data Visualization

- Missing-value heatmap of the original combined dataset.

- Box plot of numerical physicochemical features for outlier inspection.
- Count plot showing red and white wine records.
- Correlation heatmap for numerical columns.
- Histogram showing the distribution of alcohol.

7. Business Insights

The notebook calculates five dataset observations from the cleaned data: total retained records, red wine count, white wine count, average quality score, and average alcohol percentage. This avoids inventing results before the code is executed.

Three data quality checks/issues: missing-value checks and imputation; duplicate detection and removal; consistent formatting and numeric type validation.

Three techniques applied: median/mode imputation; label encoding; IQR-based outlier detection and capping.

Recommendations before model training:

- Split the data into training and testing sets before fitting model transformations.
- Use a preprocessing pipeline so imputation, scaling, and encoding are learned from training data only, avoiding data leakage.
- Check target-class balance and choose suitable evaluation metrics; quality scores may be imbalanced.

Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. The notebook and script are included in this project folder.

Important: Run the notebook before final submission and review the printed outputs and generated plots. The report deliberately does not invent post-cleaning statistics; those values are produced by the notebook.